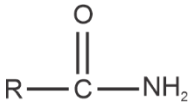
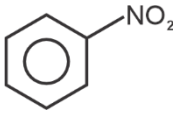
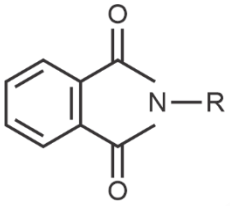
CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Match the compounds (List-I) with the appropriate Catalyst/Reagents (List-II) for their reduction into corresponding amines.

List-I	List-II
(A) 	(I) NaOH (aqueous)
(B) 	(II) H_2/Ni
(C) R–C≡N	(III) $LiAlH_4$, H_2O
(D) 	(IV) Sn, HCl

Choose the correct answer from the options given below:

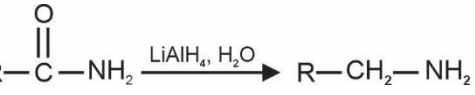
(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(2) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(3) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

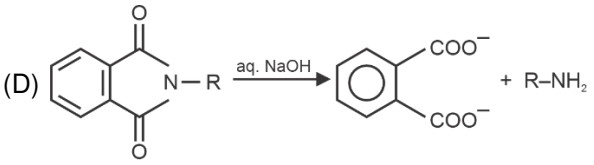
(4) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Answer (1)

Sol. (A) 

(B) 

(C) $R–C\equiv N \xrightarrow{H_2,Ni} R–CH_2–NH_2$

(D) 

52. Match **List-I** with **List-II**

List-I (Partial Derivatives)	List-II (Thermodynamic Quantity)
(A) $\left(\frac{\partial G}{\partial T}\right)_P$	(I) C_P
(B) $\left(\frac{\partial H}{\partial T}\right)_P$	(II) –S
(C) $\left(\frac{\partial G}{\partial P}\right)_T$	(III) C_V
(D) $\left(\frac{\partial U}{\partial T}\right)_V$	(IV) V

Choose the **correct** answer from the options given below:

(1) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(2) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(3) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(4) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Answer (4)

Sol. $\because$ dH = dq (at P = constant)

$dH = C_P dT$

$\left(\frac{dH}{dT}\right)_P = C_P$

dU = dq (at V = constant)

$dU = C_V dT$

$\left(\frac{dU}{dT}\right)_V = C_V$

$\because$ dG = VdP – SdT

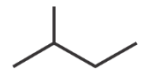
at P = constant, dP = 0



$$\left(\frac{dG}{dT}\right)_P = -S$$

at T = constant

$$\left(\frac{dG}{dP}\right)_T = V$$

53. $RBr \xrightarrow[(ii)\ H_2O]{(i)\ Mg,dry\ ether}$  2-Methylbutane

The maximum number of RBr producing 2-methylbutane by above sequence of reactions is _____. (Consider the structural isomers only)

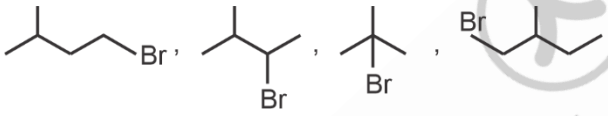
(1) 3 (2) 4

(3) 1 (4) 5

Answer (2)

Sol. $RBr \xrightarrow[dry\ Ether]{Mg} RMgBr \xrightarrow{H_2O} R-H$

Hence, RBr Can be



Total 4 Structural isomers.

54. The species which does not undergo disproportionation reaction is:

(1) ClO^-

(2) ClO_3^-

(3) ClO_2^-

(4) ClO_4^-

Answer (4)

Sol. The species having Cl-atom in its maximum oxidation state of (+7) or minimum oxidation state of (–1) will not undergo disproportionation reaction.

ClO_4^- has Cl in (+7) oxidation state.

55. Given below are two statements:

Statement (I): Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

Statement (II): The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) **Statement I** is true but **Statement II** is false

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is false but Statement II is true

Answer (2)

Sol. During corrosion metal oxidises, and acts as anode in acidic medium and impure metal as cathode.

Hence, acidic medium increases the rate of corrosion.

56. Density of 3 M NaCl solution is 1.25 g/mL. The molality of solution is

(1) 2 m (2) 3 m

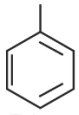
(3) 1.79 m (4) 2.79 m

Answer (4)

Sol. $\text{Molality} = \frac{1000 \times M}{1000 \times d - M.M_w}$

$$\text{Molality} = \frac{1000 \times 3}{1000 \times 1.25 - 3 \times 58.5} = \frac{3000}{1074.5}$$

= 2.79 molal

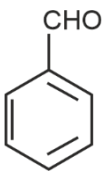
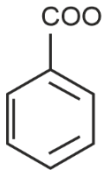
57.  Toluene (excess) $\xrightarrow[(ii)\ H_3O^+ \\ (iii)\ NaHSO_3]{(i)\ CrO_2Cl_2,CS_2}$ Filter $\longrightarrow$ Residue (A)

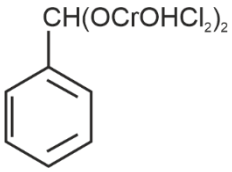
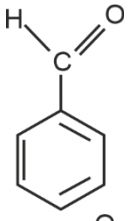
Residue (A) + HCl (dil.) $\rightarrow$ Compound (B)

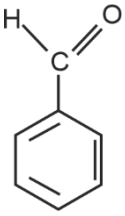
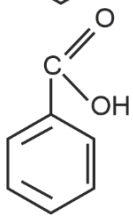
Structure of residue (A) and compound (B) formed respectively is

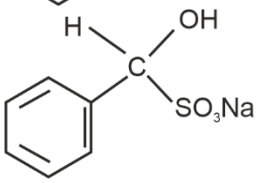
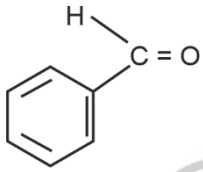
(A) **(B)**



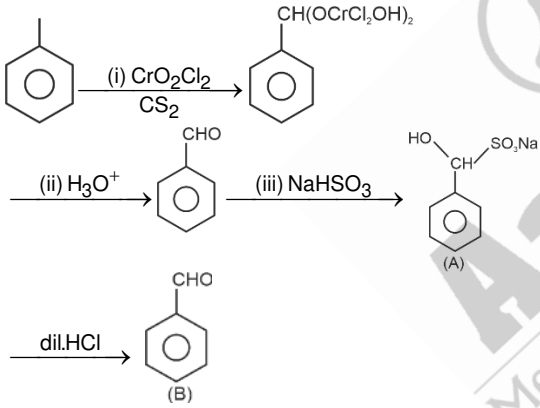
(1)  

(2)  

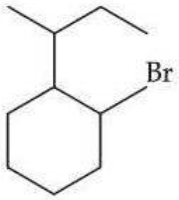
(3)  

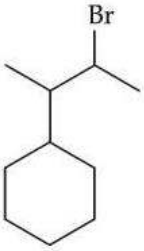
(4)  

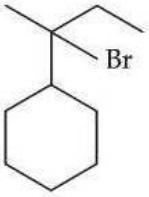
Answer (4)

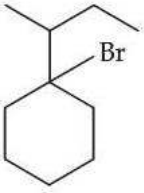
Sol. 

58. When sec-butylcyclohexane reacts with bromine in the presence of sunlight, the major product is:

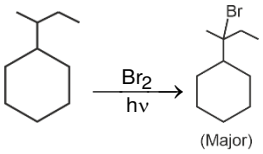
(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. 

59. Given below are two statements:

Statement (I): A spectral line will be observed for a $2p_x \rightarrow 2p_y$, transition.

Statement (II): $2p_x$ and $2p_y$ are degenerate orbitals.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) **Statement I** is true but **Statement II** is false

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is false but **Statement II** is true

Answer (4)

Sol. $2p_x$ and $2p_y$ are degenerated orbitals hence having equal energy and therefore no spectral line will be observed for $2p_x \rightarrow 2p_y$ transition.

60. Identify the homoleptic complex(es) that is/are low spin.

(A) $[Fe(CN)_5NO]^{2-}$ (B) $[CoF_6]^{3-}$

(C) $[Fe(CN)_6]^{4-}$ (D) $[Co(NH_3)_6]^{3+}$

(E) $[Cr(H_2O)_6]^{2+}$

Choose the correct answer from the options given below:

(1) (C) and (D) only (2) (C) only

(3) (B) and (E) only (2) (A) and (C) only

Answer (1)

Sol. Except $[Fe(CN)_5NO]^{2-}$ all are homoleptic as have only one type of ligand.

High spin complexes are $[CoF_6]^{3-}$, $[Cr(H_2O)_6]^{2+}$

Low spin complexes are $[Fe(CN)_6]^{4-}$, $[Co(NH_3)_6]^{3+}$

Hence (C) and (D) are homoleptic and low spin complexes.



61. The molar solubility(s) of zirconium phosphate with molecular formula $(Zr^{4+})_3(PO_4^{3-})_4$, is given by relation:

(1) $\left(\frac{K_{sp}}{8435}\right)^{\frac{1}{7}}$ (2) $\left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$

(3) $\left(\frac{K_{sp}}{5348}\right)^{\frac{1}{6}}$ (4) $\left(\frac{K_{sp}}{9612}\right)^{\frac{1}{3}}$

Answer (2)

Sol. $Zr_3(PO_4)_4(s) \rightleftharpoons \underset{3s}{3Zr^{4+}}(aq) + \underset{4s}{4PO_4^{3-}}(aq)$

$K_{sp} = (3s)^3.(4s)^4$

$K_{sp} = 6912\, s^7$

$s = \left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$

62. The alkane from below having two secondary hydrogens is:

(1) 2,2,3,3-Tetramethylpentane

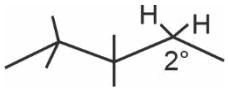
(2) 2,2,4,4-Tetramethylhexane

(3) 4-Ethyl-3,4-dimethyloctane

(4) 2,2,4,5-Tetramethylheptane

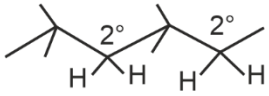
Answer (1)

Sol. 2,2,3,3-Tetramethylpentane



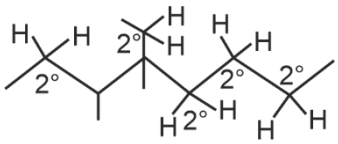
2 secondary Hydrogen

2,2,4,4-Tetramethylhexane



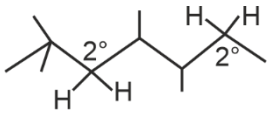
4 secondary Hydrogen

4-Ethyl-3,4-dimethyloctane



10 secondary Hydrogen

2,2,4,5-Tetramethylheptane



4 secondary Hydrogen

63. The correct order of the following complexes in terms of their crystal field stabilization energies is:

(1) $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+}$

(2) $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+}$

(3) $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_4]^{2+}$

(4) $[Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_4]^{2+} < [Co(en)_3]^{3+}$

Answer (2)

Sol. Crystal field splitting energy $(\Delta) \propto$ charge or oxidation state of central metal atom.

Crystal field splitting energy $(\Delta) \propto$ Field strength of ligand (and chelation)

Crystal field stabilisation energy (CFSE) = $[-0.4\, t_{2g} + 0.6\, eg]\Delta_o$ (for octahedral)

For,

$\left[Co(en)_3\right]^{3+} : Co^{3+} : t_{2g}^6 eg^0 ; CFSE = -2.4(\Delta_0)_1$

$\left[Co(NH_3)_6\right]^{3+} : Co^{3+} : t_{2g}^6 eg^0 ; CFSE = -2.4(\Delta_0)_2$

$\left[Co(NH_3)_6\right]^{2+} : Co^{2+} : t_{2g}^5 eg^2 ; CFSE = -0.8(\Delta_0)_3$

$\left[Co(NH_3)_4\right]^{2+} : Co^{2+} : e^4 t_2^3 ; CFSE = -1.2\Delta_t$

$\left(as : (\Delta_t) = \frac{4}{9}(\Delta_0)_3\right)$

$\because \quad \Delta_t < (\Delta_0)_3 < (\Delta_0)_2 < (\Delta_0)_1$



64. Given below are two statements:

Statement (I): Nitrogen, sulphur, halogen and phosphorus present in an organic compound are detected by Lassaigne's Test.

Statement (II): The elements present in the compound are converted from covalent form into ionic form by fusing the compound with Magnesium in Lassaigne's test.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true

(4) **Statement I** is true but **Statement II** is false

Answer (4)

Sol. For Lassaigne's test, sodium is used and not magnesium to convert covalent to ionic form.

65. Given below are two statements:

Statement (I): An element in the extreme left of the periodic table forms acidic oxides.

Statement (II): Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

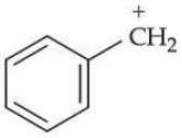
(3) Statement I is false but Statement II is true

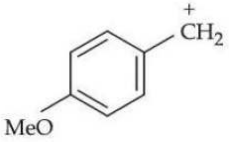
(4) Statement I is true but Statement II is false

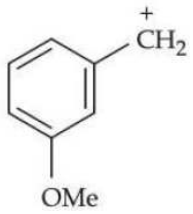
Answer (3)

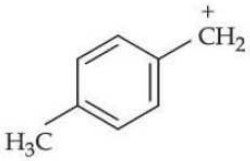
Sol. Group I elements forms basic oxides. Group 17 elements forms acids with their oxides on reaction with water.

66. The most stable carbocation from the following is:

(1) 

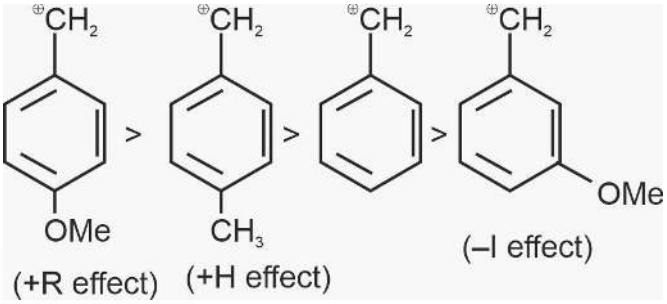
(2) 

(3) 

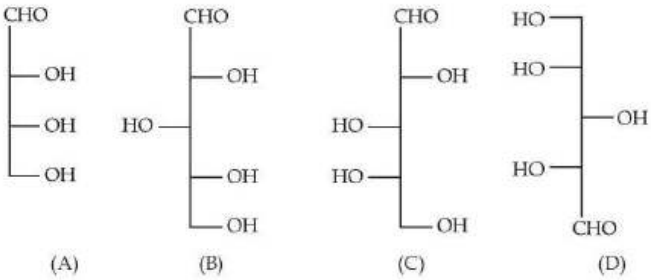
(4) 

Answer (2)

Sol. The order of stability of carbocations given is:



67. Identify the number of structure/s from the following which can be correlated to D-glyceraldehyde



(1) four

(2) three

(3) one

(4) two

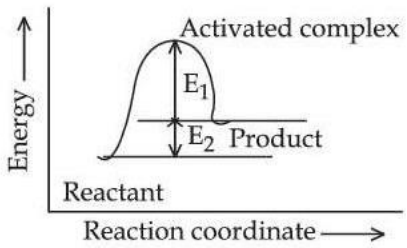
Answer (2)

Sol. For D-configuration, when -CHO is at top, -OH at the lowest chiral center should be on right side.

Hence, (A), (B) and (D) can be correlated to D-glyceraldehyde.



68. Consider the given figure and choose the correct option



(1) Activation energy of backward reaction is E_1 and product is more stable than reactant

(2) Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant

(3) Activation energy of forward reaction is $E_1 + E_2$ and product is more stable than reactant

(4) Activation energy of both forward and backward reaction is $E_1 + E_2$ and reactant is more stable than product

Answer (2)

Sol. E_1 : Activation energy for backward reaction.

$E_1 + E_2$: Activation Energy for forward reaction.

Product has more energy than reactant

69. Arrange the following compounds in increasing order of their dipole moment:

HBr, H_2S, NF_3 and $CHCl_3$

(1) $NF_3 < HBr < H_2S < CHCl_3$

(2) $H_2S < HBr < NF_3 < CHCl_3$

(3) $HBr < H_2S < NF_3 < CHCl_3$

(4) $CHCl_3 < NF_3 < HBr < H_2S$

Answer (1)

Sol. $\mu_{HBr} = 0.78$ D

$\mu_{H_2S} = 0.95$ D

$\mu_{NF_3} = 0.24$ D

$\mu_{CHCl_3} = 1.01$ D

Hence dipole moment of

$NF_3 < HBr < H_2S < CHCl_3$

70. The maximum covalency of a non-metallic group 15 element 'E' with weakest E – E bond is

(1) 6 (2) 3

(3) 5 (4) 4

Answer (4)

Sol. E-E bond strength decreases down the group 15. But N-N bond is weakest due to e^- repulsion

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. Consider the following cases of standard enthalpy of reaction $\left(\Delta H_r^\circ \text{ in kJ mol}^{-1}\right)$

$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)\ \Delta H_1^\circ$

$= -1550$

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)\ \Delta H_2^\circ = -393.5$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)\ \Delta H_3^\circ = -286$

The magnitude of $\Delta H_{f\,C_2H_6(g)}^\circ$ is _____ kJ mol^{-1} (Nearest integer).

Answer (95)

Sol. $C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)\ \Delta H_1^\circ$

$= -1550$ …(i)

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)\ \Delta H_2^\circ = -393.5$ …(ii)

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)\ \Delta H_3^\circ = -286$ …(iii)

From 2 × eq^n(ii) + 3 × eq^n(iii) – eq^n(i)

$2C_{(\text{graphite})} + 3H_2(g) \rightarrow C_2H_6(g) : \Delta H_{f\,C_2H_6}^\circ$

$\left(\Delta H_f^\circ\right)_{C_2H_6} = 2\times(-393.5) + 3\times(-286) - (-1550)$

$= -95$ kJ/mol



72. The compound with molecular formula C_6H_6, which gives only one monobromo derivative and takes up four moles of hydrogen per mole for complete hydrogenation has _____ π electrons.

Answer (8)

Sol. Since 4 moles of H_2 is being added for complete hydrogenation the degree of unsaturation = 4

No. of π electrons in $C_6H_6 = 4 \times 2 = 8$

73. Niobium (Nb) and ruthenium (Ru) have "x" and "y" number of electrons in their respective 4*d* orbitals. The value of x + y is _____.

Answer (11)

Sol. Ru : $[Kr]4d^7 5s^1$; Nb : $[Kb]4d^4 5s^1$

x = 7, y = 4

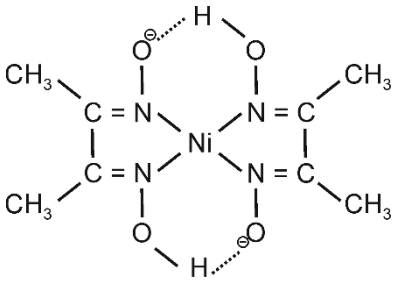
x + y = 11

74. The complex of Ni^{2+} ion and dimethyl glyoxime contains _____ number of Hydrogen (H) atoms.

Answer (14)

Sol. Ni^{2+} with(dmg) forms $[Ni(dmg)_2]^{2+}$ having 2 H-Bonds as shown:

The no. of H atoms = 14



75. 20 mL of 2 M NaOH solution is added to 400 mL of 0.5 M NaOH solution. The final concentration of the solution is _____ $\times 10^{-2}$ M. (Nearest integer)

Answer (57)

Sol. $[NaOH]_{final} = \frac{20 \times 2 + 400 \times 0.5}{420} = \frac{40 + 200}{420} = \frac{240}{420}$

= 0.57 M

$[NaOH]_{final} = 57 \times 10^{-2}$ M

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